

Game 01 — Tap or Don't Tap (Complete Plan)

Game 1 of 7 Complexity: ★ Low Estimate: 1–2 days Status: Mostly built

Complete plan (supersedes the 2026-09-09 sample). Status: **built & browser-verified** — re-synced with the implementation 2026-09-13 (deviations recorded in §13.1 below). (2026-09-09 build pass) — `apps/frontend/public/games/tap-or-dont-tap/` with decoys, Stroop traps, rule swap, audio, pause and a baked percentile table. Logic tests: `apps/frontend/src/__tests__/games-tap-or-dont-tap.test.ts` .

****Rev 2 (2026-09-11): architecture upgraded — pure `core.js` , `config.js` , `storage.js` (versioned + migrated), `audio.js` ; analytics removed; a11y pass. 30/30 tests green.**** This doc is the spec of record: the constants below are authoritative when tuning, and §13 tracks the Rev 2 state.

1. Overview

A Go/No-Go reaction game (the real psychology self-control test). The shareable number is the **best reaction time in ms**. Success metric: median first session ≥ 15 rounds; players can state their ms score from memory after one session.

2. Complete game spec

Round loop (state machine: `MENU → WAITING → SIGNAL → FEEDBACK → (WAITING | GAMEOVER)` , plus `PAUSED`):

- WAITING** — dark screen, random delay **uniform 300–3000 ms**. A tap here is a **false start**: lose a heart, skip the round (no signal shown).
- SIGNAL** — one of:
 - GREEN → tap within the window.
 - RED → do not tap; survives after the window expires.
 - ● / ● DECOY (round ≥ 10) → never tap; ignoring it for the full window = +50.
 - STROOP TRAP (round ≥ 15) — word "TAP" in red or "WAIT" in green; **color wins, word ignored**.
- FEEDBACK** — 400 ms overlay ("218ms! 🔥 " / " ❌ Too slow " / " ✅ Resisted +25"), then the next round begins. (Amended 2026-09-12 to match shipped tuning — no blank gap.)

Constants (authoritative — implemented in `game.js`)

Constant	Value	Notes
<code>WAIT_MIN/MAX_MS</code>	300 / 3000	uniform random
<code>WINDOW_START_MS</code>	1200	signal visible duration, round 1
<code>WINDOW_SHRINK_MS</code>	25	per round

Constant	Value	Notes
<code>WINDOW_FLOOR_MS</code>	450	never below
<code>HEARTS</code>	3	0 → game over
<code>GREEN_BASE_POINTS</code>	100	<code>points = min(100 + streak×10, 500)</code> (×5 cap)
<code>RESIST_BONUS</code>	25	red expired untouched
<code>DECOY_BONUS</code>	50	decoy expired untouched (round ≥ 10)
<code>FEEDBACK_MS</code>	400	overlay dwell, then next round (amended 2026-09-12)
<code>SWAP_FROM_ROUND</code> / <code>SWAP_CHANCE</code>	12 / 0.15	from round 12, 15 %/round; swap lasts <code>SWAP_ROUNDS</code> = 3 rounds — "📄 RULES SWAPPED!" (amended 2026-09-12 to match shipped tuning)

Signal mix per round band (amended 2026-09-12 to match shipped tuning): 1–9 → 55 % green / 45 % red; 10–14 → 45 green / 35 red / 20 decoy; 15+ → 40 green / 30 red / 15 decoy / 15 Stroop.

Losing: tap on red/decoy/Stroop-wrong, false start, or green missed → –1 heart each.
Reaction recorded only on green hits. Swap inverts green/red only; decoys always "never tap".

3. Progression & percentile model

- Difficulty = shrinking window + signal mix. No other knobs.
- **Best reaction** = min green-hit reaction of the run; **score** = sum of points.
- **Top-% badge**: `BAKED_TABLE` in `game.js` maps score → percentile (fixed simulated distribution). Personal history (last 20 runs `{score, bestMs}`) stored locally (amended 2026-09-12: cap 20, no per-entry date — matches shipped schema).

4. Screens & UI

State	Elements
<code>menu</code>	Title, one-line rule, best score + best ms, ▶ Start, mute toggle
<code>waiting</code>	Full-dark screen, "wait for it..." hint, HUD (hearts, score, round)
<code>signal</code>	Full-bleed color, optional overlay word (Stroop), HUD persists
<code>feedback</code>	Center overlay text + tone color, 400 ms
<code>gameover</code>	Card: score, best ms, rounds, top-% badge, 📄 Share, ↺ Retry, △ Menu
<code>paused</code>	Overlay "Paused — tap to resume" (auto on <code>visibilitychange</code>)

Whole screen is the tap target (`pointerdown`); `touch-action:none` on the play area;
Space/Enter equivalent on desktop; HUD height fixed across states (no relayout).

5. Data model (localStorage, guarded wrapper)

```
game:tap-or-dont-tap:best    → { score: 1240, bestMs: 187 }
game:tap-or-dont-tap:history → [{ score: 1240, bestMs: 187 }, ...max 20]
game:tap-or-dont-tap:muted   → true|false
```

6. File structure & function inventory

```
tap-or-dont-tap/
  index.html    # state sections + gameover card
  style.css     # flash colors, HUD, overlays
  game.js       # constants → pure core → state machine → DOM wiring
```

Pure core (no DOM — the test surface, currently in `game.js`, mirrored in the site test

file): `windowMs(round)`, `pointsForGreen(streak)`, `pickSignal(round, rng)`,
`resolveTap({signalKind, expectsTap, tapped, elapsedMs})`, `percentileFor(score)`.

DOM layer: `startRun`, `beginRound`, `showSignal`, `handleTap`, `expireSignal`,
`finishRound`, `loseHeart`, `gameOver`, `pauseRun`, `showFeedback`, audio helpers.

7. Edge cases (verify each once)

1. Hidden tab during WAITING → pause; on return restart the wait from 0. Hidden during SIGNAL → counts as a miss (anti-cheat).
2. Tap within 50 ms of a state change → single tap only (pointerdown debounce).
3. Private mode / storage disabled → game runs, persistence silently skipped.
4. Phone sleep mid-wait → `visibilitychange` → pause path.
5. Stroop × swap overlap (rounds 21–23) → swap inverts green/red; decoys never tappable.
6. Rapid restart from gameover → all pending timers cleared.
7. Reaction ≤ 0 ms or $>$ window + 50 ms → discarded as anomaly (clock sanity).

8. Testing plan

Existing: `apps/frontend/src/__tests__/games-tap-or-dont-tap.test.ts` — keep green.

Required cases (verify present; add missing): `windowMs` ramp incl. floor;

`pointsForGreen` cap; `resolveTap` truth table (5 signal kinds × tap/late/no-tap);

seeded `pickSignal` sequences per band; `percentileFor` monotonic + bounds.

9. Task breakdown

P0 — playable core — DONE (2026-09-09 build pass)

- ☒ State machine + round loop; signal kinds; hearts; reaction capture; ramp; scoring
- ☒ HUD + gameover card; pause/resume (`visibilitychange` incl. wait-restart)

P1 — full rules & feel — DONE

- ✓ Streak multiplier; decoys; Stroop traps; rule swap; false starts
- ✓ Feedback overlays; WebAudio + mute; baked percentile badge

P2 — persistence/share/QA

- ✓ Best score + history persistence; share chain
- ✓ **Delete** `analyticsCandidates` / `trackGamePlayed` **stubs** — done 2026-09-11 (Rev 2); the game makes zero network calls (isolation greps clean)
- ✓ Storage/audio/config helpers moved to per-game modules (`storage.js` , `audio.js` , `config.js`) per the Rev 2 architecture — supersedes the `../shared/` idea
- ✓ Pure core extracted to `core.js` ; site test file re-pointed and green (30 tests)
- ✓ QA gate (master README §5): offline play + isolation greps clean (2026-09-11)

P3 — polish

- ✓ History sparkline on menu; haptics on heart loss; sound design pass (sparkline: `core.js` `sparklinePoints` + menu SVG; haptics: `audio.js` `buzz()`) — verified 2026-09-13; the checkbox was stale)
- ☐ Reduced-motion variant of flash (shipped: CSS disables feedback animation and button transitions under `prefers-reduced-motion`); tuning pass after 5 human test sessions still owed

10. Acceptance criteria

- §7 edge cases each verified once (note date when done); §8 tests green.
- 20 consecutive rounds, no stuck state; restart from gameover < 300 ms to first signal.
- Share works through all three fallback paths (share sheet / clipboard / prompt).

11. Deferred coupling (intentionally NOT built)

Footer/nav links, `game_played` analytics posting, achievements, leaderboard backend — per master README §7. The analytics stubs in `game.js` are to be **deleted**, not wired.

12. Remaining work (from code review 2026-09-10)

1. ~~Remove analytics stubs~~ — done 2026-09-11 (Rev 2 R2-1; isolation greps clean).
2. ~~Shared-utils migration + `core.js` extraction~~ — done 2026-09-11 as per-game `config.js` / `storage.js` / `audio.js` + `core.js` (Rev 2 R2-1; supersedes `../shared/`).
3. ~~Reduced-motion + tuning~~ — reduced-motion covered (flash is a solid color; pop + button transitions disabled); tuning pass still owes human sessions (P3).

13. Rev 2 architecture upgrade (2026-09-11) — reference: `03-sliding-puzzle.md`

Owner-approved architecture reference for all games (Sliding Puzzle Rev 2), applied per this game's nature: a reflex game has no levels and no meaningful multiplayer, so

the upgrade here is **hygiene + structure**, not features. Isolation contract unchanged: no analytics, no site coupling, offline-first.

Target file layout

```
tap-or-dont-tap/
  index.html
  style.css
  config.js      # flags + per-locale strings (menu rule line, feedback/share/gameover
                 # copy), host-overridable: window.__TAP_OR_DONT_TAP_CONFIG__ → ?locale= → defaults
  storage.js     # guarded facade: versioned save + migrations, remote adapter slot
  audio.js       # WebAudio + mute
  core.js        # extracted pure core (windowMs, pointsForGreen, pickSignal, resolveTap,
                 # percentileFor) – the test surface; jest suite re-pointed here
  game.js        # UI shell: state machine + DOM wiring
```

config.js

Flags and strings only — no multiplayer, no levels in this game. Per-locale strings for the rule line, feedback overlays, share template and gameover card (shipped 2026-09-12 — 18 keys, all consumed); `prefs` may override the copy but is documented as a cache, not the source of truth.

storage.js facade

One versioned save record replaces the three loose keys:

```
game:tap-or-dont-tap:save → { version:1, best:{score,bestMs}, history:[...≤50], prefs:{muted} }
```

with a one-time migration from the legacy keys. Remote adapter slot (host-injected only) reserved for future account sync of best/history — the game never checks auth; ****guests keep full local persistence****. Mid-round resume: N/A (rounds are seconds).

Multiplayer (per this game's nature)

- Live/real-time duels: **gated** — device input/display latency makes reaction times unfairly comparable; needs a backend plus server-side timing and a \$7 reversal. Not planned.
- Optional future: **async seeded duel** — `pickSignal(round, rng)` already accepts an `rng`, so a shared seed fixes the identical signal sequence for both players; compare score. Deferred; not scheduled.

Phases

-  R2-1 Hygiene (2026-09-11): analytics stubs deleted; pure core extracted to `core.js` (jest suite re-pointed); `config.js` (locale + share strings, host-overridable) and `storage.js` (versioned `save` document, legacy-key migration, remote adapter slot) introduced; `audio.js` split out. 30/30 tests green; browser-verified end-to-end.

- ☒ R2-2 A11y/polish (2026-09-11): signal `aria-label` describing each flash (incl. the Stroop lie) via `role="img"`; focus lands on Retry at gameover; `data-state` now maintained on the game root; Stroop word contrast fixed on the green flash; go-badge gradient darkened to AA; visible `:focus-visible` rings. (Reduced-motion: the flash is a solid color — nothing moves; the existing block covers the feedback pop + buttons. Sparkline + haptics were already shipped.)
- ☒ R2-1b Cleanup (2026-09-12, stale-code scan): unused `isMuted` / `whenSec` removed; history writes trimmed to the plan schema (`{score, bestMs}` — `rounds` / `ts` dropped); dead flash-yellow word rule removed; `defaultSave()` single-sourced; config strings completed (menu rule, swap banner, feedback overlays, gameover card); `FEEDBACK_MS` single-sourced. 31/31 tests green.
- ☐ R2-3 (deferred, optional) async seeded duel — only if there's demand. Still not built; `generateRound`'s injectable rng is the only seam kept for it.

13.1 Implementation sync (2026-09-13) — deviations of code from the text above

Recorded verbatim from a code audit; the plan text above is otherwise accurate.

- **Function inventory drift (\$6/\$13):** shipped names are `windowMsForRound`, `generateRound`, `resolveRound(spec, opts)`, `bakedPercentile` / `localPercentile` (not `windowMs`, `pickSignal`, `resolveTap`, `percentileFor`). Extras: `effectiveExpectsTap` (Stroop lie) and `sparklinePoints`.
- **Badge = hybrid, not baked-only (\$3):** with ≥ 5 recorded runs the top-% badge comes from the player's own history (`localPercentile`); `BAKED_TABLE` is the fallback. The table lives in `core.js`, not `game.js`.
- **History cap is 20**, not \$13's " ≤ 50 " (\$3's amendment is the accurate one).
- **No distinct PAUSED state (\$2):** the shell keeps waiting/signal states plus a paused-overlay flag; hidden-during-signal pauses and restarts the round on resume instead of scoring a miss (kinder than \$7.1's "counts as a miss").
- **No 50 ms tap debounce and no >window+50 ms anomaly discard (\$7.2/\$7.7):** the feedback state gates double-taps instead; reactions clamp to ≥ 1 ms only.
- **Share chain lacks the third `prompt()` fallback (\$10):** Web Share → clipboard ("Copied" note) → clipboard failure is swallowed silently.
- `?debug=1` **does not exist in this game** (never spec'ed here; `?locale=` only).
- **Back-link exists:** ← All games to `/games` — \$11's "no nav links" referred to the product footer/nav; the hub back-link ships in every game.
- **No in-folder test.html:** tests are jest-only (31 cases).

Enhancements pass — shipped 2026-09-13, verified 2026-09-15

Folded 2026-09-16 from `plan/suggestion/01` (retired; full original task text in git history).

Three user-friendliness upgrades, all shipped (commit `7eb296f`): colorblind shape cues on every flash (shape follows the actual color, never the word — grayscale-safe by design), one-time tutorial toasts for decoy/Stroop/rule-swap via save v2 (`prefs.seenTutorials`, v1→v2 migration), and distinct loss-reason copy (`missedIt` / `wasRed` / `wasDecoy` / `stroopTrap`; the old `tooSlow` / `stroopLie` keys were replaced). The "Too early!" false-start copy predates the pass

and was kept. Verified 2026-09-15: live browser play confirmed the decoy tutorial toast fires and persists (`seenTutorials.decoy: true` in a real save v2) and the "Missed it!" loss-reason overlay renders; decoy generation (rounds 10+, yellow/blue, never-tap), rule-flip inversion, decoy +50 scoring, and the v1→v2 tutorial-flag migration are pinned by `games-tap-or-dont-tap.test.ts` (all 8 games suites green, 343 tests).

Source: plan/games markdown · generated 2026-09-09 · part of the 2D Games sample plans (plan/games/pdf).